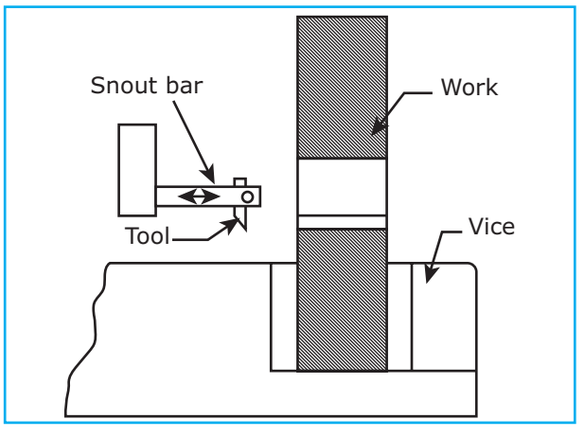
Machining internal keyways

Internal keyways are cut inside the holes of gears and pulleys. It is done by holding the tool on a special tool holder called 'snout bar'. The snout bar is directly fitted on the clapper block.

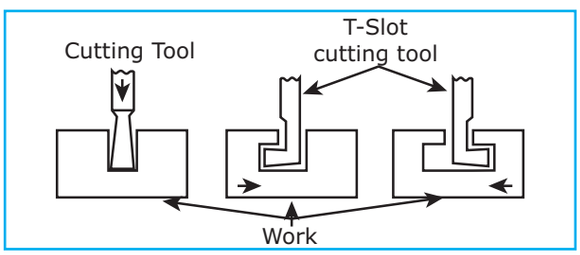
Illustrate machining a external keyway and a internal keyway respectively.



Machining internal keyways

T-slot machining

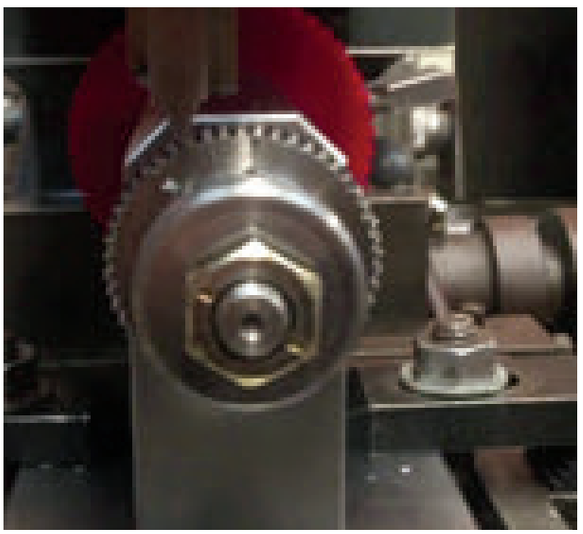
The shape of the T-slot is marked on the face of the work. A parting off tool is fitted on the toolpost and a rectangular slot is machined at the middle for the required depth. The broad base of the 'T' slot is machined by a T-slot cutting tool. Figure illustrates machining a 'T' slot.



'T' slot cutting

Machining a rack gear

Rack gear cutting is a process of cutting teeth elements at linear pitch on a flat piece of work. Firstly, the groove is machined with a square nose parting tool. Then, the groove is further machined with a form tool confirming the shape of the teeth.



Machining Rackgear

3.16 Cutting speed, depth of cut and Feed

Cutting speed

The distance an object travels in a particular period of time is known as speed. In a shaper, the cutting speed is the speed at which the metal is removed by the cutting tool in a period of one minute. In a shaper, the cutting speed is considered only during the forward cutting stroke. This is expressed in metre per minute.

The cutting speed differs to suit different machining conditions like work material, the finish required, and the type of the tool and the rigidity of the machine.

Depth of cut

Depth of cut is the thickness of metal that is removed during machining. It is the perpendicular distance measured between the machined surface and the uncut surface of the workpiece. It is expressed in mm or in inches.

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